PHYSICS - ELECTROSTATIC POTENTIAL AND CAPACITANCE

Chapter 2 | Objective Questions | Class 12 | Maximum Time: 120 minutes

खण्ड अ — वस्तुनिष्ठ प्रश्न (Objective Type)

1. If a uniform electric field exists along the X-axis, then the equi-potential surface will be
(A) anywhere (B) in the YZ-plane (C) in the XY-plane (D) in the XZ-plane

2. Work done in moving a charge between two points in an electrostatic field does not depend on the:
(A) path taken (B) value of the charge (C) potential difference (D) shape of the conductor

3. On inserting a dielectric material between the plates of a parallel plate capacitor, its capacitance
(A) decreases (B) increases (C) remains unchanged (D) cannot be said

4. For an isolated conductor, which of the following ratios is constant?
(A) potential difference/total charge (B) None of these
(C) $\frac{\text{Total charge}}{\text{Potential}}$ (D) $\frac{(\text{Total charge})^2}{\text{Potential}}$

5. The work done by the electric force in moving a charge along an equipotential surface is:
(A) infinite (B) zero (C) maximum (D) negative

6. Statement-I: The flow of negative charge is always from lower potential to higher potential. Statement-II: Electric potential is a vector quantity.
(A) Statement-I is correct, but Statement-II is false.
(B) Both statements are correct but Statement-II is not the correct explanation of Statement-I.
(C) Both statements are correct and Statement-II is the correct explanation of Statement-I.
(D) Statement-I is false, but Statement-II is correct.

7. An electrostatic field is—
(A) Both (A) and (B) (B) Conservative (C) None of these (D) Non-conservative

8. Potential in the direction of the electric field:
(A) first increases, then decreases (B) remains constant
(C) increases (D) decreases

9. Plate X of a 2 μF capacitor is at 10 V and plate Y is at 20 V. The charge on plate X is—
(A) $-10\mu C$ (B) $-20\mu C$ (C) Zero (D) $20\mu C$

10. In a uniform electric field E, the magnitude of potential difference over distance d along the field is:
(A) $E + d$ (B) d/E (C) Ed (D) E/d

11. Statement-I: The capacitance of a capacitor increases with the increase of charge on its plate. Statement-II: $V \propto Q$
(A) Statement-I is incorrect, but Statement-II is correct.
(B) Both Statement-I and Statement-II are correct, and Statement-II is the correct explanation of Statement-I.
(C) Both Statement-I and Statement-II are correct, but Statement-II is not the correct explanation of Statement-I.
(D) Statement-I is correct, but Statement-II is incorrect.

12. Energy density of electric field
(A) 3.6×10^6 J (B) Reduces alternating current
(C) Infinite (D) $\frac{1}{2}\epsilon_0 E^2$

13. To find the total potential at a point due to two point charges, their potentials are:
(A) added algebraically (B) multiplied
(C) compared only by magnitude (D) added as vectors

14. Three capacitors, each of 4 μF, are connected in series between A and B. The equivalent capacitance is—
(A) $2/3\mu F$ (B) $4\mu F$ (C) $3\mu F$ (D) $4/3\mu F$

15. 64 identical drops, each of capacitance 5 μ F, combine to form a bigger drop. What will be the capacitance of the bigger drop?
(A) 4 μF (B) 164 μF (C) 20 μF (D) 25 μF

16. At a distant point, the potential due to an electric dipole is inversely proportional to:
(A) r^2 (B) $\sqrt{r}$ (C) r^3 (D) r

17. In the circuit, 3 μF and 2 μF capacitors are in parallel, and this combination is in series with a 4 μF capacitor. The equivalent capacitance between A and B is—
(A) 1 μF (B) 20/9 μF (C) 9 μF (D) 1/9 μF

18. Statement-I: The upper end of a lightning conductor installed on the roof of a building is pointed. Statement-II: This is done to increase the surface charge density.
(A) Statement-I is incorrect, but Statement-II is correct.
(B) Statement-I is correct, but Statement-II is incorrect.
(C) Both statements are correct and Statement-II is the correct explanation of Statement-I.
(D) Both statements are correct but Statement-II is not the correct explanation of Statement-I.

19. When dipole moment p is aligned parallel to a uniform electric field E, its potential energy is:
(A) $-pE$ (B) $+pE$ (C) p/E (D) zero

20. 64 identical drops, each of capacitance $5\mu F$, combine to form a big drop. What will be the capacitance of the big drop?
(A) $4\mu F$ (B) $25\mu F$ (C) $20\mu F$ (D) $164\mu F$

21. The intensity of the electric field at any point on the surface of a charged conductor is:
(A) Is perpendicular to the surface (B) Is zero
(C) Is at 45° to the surface (D) Is tangential to the surface

22. The magnitude of torque on a dipole in a uniform electric field is:
(A) p/E (B) $pE\sin\theta$ (C) E/p (D) $pE\cos\theta$

23. In the figure, two 6 μF capacitors in series form one branch, in parallel with a single 6 μF capacitor. The equivalent capacitance between A and B is—
(A) $1/9\mu F$ (B) $18\mu F$ (C) $6\mu F$ (D) $9\mu F$

24. 64 identical drops, each of capacitance $5\mu F$, combine to form a bigger drop. What will be the capacitance of the bigger drop?
(A) $164\mu F$ (B) $25\mu F$ (C) $20\mu F$ (D) $4\mu F$

25. The total work around a closed path in an electrostatic field is:
(A) positive (B) path-dependent (C) zero (D) negative

26. Three capacitors, each of capacitance C, are connected in series. The resultant capacitance will be
(A) 3C (B) 3/C (C) 1/3C (D) C/3

27. Two capacitors $C_1 = 2\mu F$ and $C_2 = 4\mu F$ are connected in series and a potential difference (p.d.) of 1200V is applied across it. The Potential difference across $2\mu F$ will be:
(A) 400V (B) 600V (C) 900V (D) 800V

28. In an electrostatic field, potential difference between two points is defined as:
(A) force per unit distance (B) work done per unit charge
(C) product of charge and distance (D) work done per unit time

29. The electric potential due to a small electric dipole at a large distance r from the center of the dipole is proportional to:
(A) $\frac{1}{r^3}$ (B) r (C) $\frac{1}{r^2}$ (D) $\frac{1}{r}$

30. Minimum number of capacitors of $2\mu F$ each required to obtain a capacitance of $5\mu F$ will be:
(A) 4 (B) 5 (C) 3 (D) 6

31. The electric field inside a charged conductor in electrostatic equilibrium is:
(A) maximum normal to the surface (B) parallel to the surface
(C) infinite (D) zero

32. Two point charges of $+10\mu C$ and $-10\mu C$ are placed at a distance 40 cm in air. Potential energy of the system will be -
(A) -2.25 J (B) 2.35 J (C) -2.35 J (D) 2.25 J

33. In electrostatic equilibrium, excess charge on a conductor resides:
(A) uniformly in its volume (B) only on its inner surface
(C) on its surface (D) only at its centre

34. The electric potential in equatorial position of an electric dipole is -
(A) $\frac{1}{4\pi\epsilon_0}\frac{p}{r}$ (B) $\frac{1}{4\pi\epsilon_0}\frac{p}{r^2}$
(C) Zero (D) $\frac{1}{4\pi\epsilon_0}\frac{p\cos\theta}{r^2}$

35. The stored energy of a capacitor charged to 100 V, is 1 J. Capacitance of the capacitor is -
(A) $2 \times 10^{-2} F$ (B) $2 \times 10^{-4} F$ (C) $2 \times 10^4 F$ (D) $2 \times 10^2 F$

36. The magnitude of electric field just outside a conductor surface is:
(A) $2\sigma/\epsilon_0$ (B) $\sigma\epsilon_0$ (C) σ/ϵ_0 (D) ϵ_0/σ

37. Charge on a capacitor of capacity $1\mu F$ connected to a source of 1 volt will be
(A) 1C (B) $1\mu C$ (C) Infinite (D) Zero

38. A capacitor of capacitance $50\mu F$ is charged with 10V. The energy of the capacitor is?
(A) $5 \times 10^{-2} J$ (B) $2.5 \times 10^{-3} J$ (C) $2.5 \times 10^{-4} J$ (D) $1.2 \times 10^{-8} J$

39. The surface charge density near a pointed part of a conductor is generally:
(A) smaller (B) zero (C) larger (D) unchanged

40. Van de Graff generator is a machine which generates
(A) High Current (B) High Voltage
(C) Both high current and high voltage (D) Only Low current

41. 64 identical drops each of capacity $5\mu F$ combine to form a big drop. Capacity of the big drop will be
(A) $4\mu F$ (B) $25\mu F$ (C) $164\mu F$ (D) $20\mu F$

42. Electrostatic shielding is based on the fact that inside a closed conducting shell:
(A) potential is infinite (B) charge cannot be nonzero
(C) field depends only on time (D) electric field is zero

43. In the circuit, 3 μF and 2 μF capacitors are in parallel, and this combination is in series with a 4 μF capacitor. The equivalent capacitance between A and B is—
(A) $1\mu F$ (B) $9\mu F$ (C) $\frac{20}{9}\mu F$ (D) $\frac{1}{9}\mu F$

44. The value of electric potential at a distance r from a point charge q is
(A) $\frac{q.r}{4\pi\epsilon_0}$ (B) $\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$
(C) $\frac{1}{4\pi\epsilon_0}\frac{q^2}{r}$ (D) $\frac{1}{4\pi\epsilon_0}\frac{q}{r}$

45. The capacitance of an isolated conductor depends on its:
(A) potential (B) shape and surroundings (C) charge (D) change in charge-to-potential ratio

46. The dielectric constant of water is
(A) 1 (B) 80 (C) 60 (D) 42.5

47. Electrical capacitance of earth of radius R is
(A) $4\pi\epsilon_0 R$ (B) $\frac{4\pi\epsilon_0}{R}$ (C) $\frac{R}{4\pi\epsilon_0}$ (D) $4\pi\epsilon_0 . R^2$

48. If the plate separation of a parallel-plate capacitor is doubled, with other conditions unchanged, its capacitance:
(A) remains unchanged (B) becomes four times (C) doubles (D) halves

49. Electric-potential is equal to
(A) wq (B) $\frac{q}{w}$ (C) $\frac{w}{q}$ (D) $\sqrt{wq}$

50. If the plate area of a parallel-plate capacitor is doubled, its capacitance:
(A) becomes four times (B) doubles (C) remains unchanged (D) halves

51. One farad is equal to
(A) 1 CV^{-1} (B) 1 CV (C) 1 CV^2 (D) 1 CV^{-2}

52. Potential energy of a charged conductor is
(A) CV^2 (B) $\frac{1}{2}CV^2$ (C) $\frac{1}{4}CV^2$ (D) $\frac{1}{3}CV^2$

53. In a series combination of capacitors, each capacitor has the same:
(A) magnitude of charge (B) potential difference (C) energy (D) capacitance

54. The value of ϵ_r in air is
(A) 1 (B) zero (C) 9×10^9 (D) infinity

55. Two capacitors, each of capacitance C, are connected in series. Their equivalent capacitance is—
(A) 2C (B) $\frac{2}{C}$ (C) $\frac{C}{2}$ (D) C

56. In a parallel combination of capacitors, each capacitor has the same:
(A) potential difference (B) charge (C) energy (D) plate area

57. Potential gradient is equal to
(A) $dx.dV$ (B) $\frac{dV}{dx}$ (C) $\frac{dx}{dV}$ (D) None of these

58. Which of the following is correct for the capacity of a parallel plate capacitor ?
(A) $\frac{\epsilon_0 d}{A}$ (B) $\frac{\epsilon_0 A}{d}$ (C) $\frac{d}{\epsilon_0 A}$ (D) $\frac{A}{\epsilon_0 d}$

59. Capacitors of 2 µF and 3 µF are connected in parallel. Their equivalent capacitance is:
(A) 1.2 µF (B) 2.5 µF (C) 6 µF (D) 5 µF

60. Dimensional formula of R.C. is
(A) $M^0L^0T^0$ (B) $M^0L^0T^{-2}$ (C) M^0L^0T (D) $M^0L^0T^{-1}$

61. Capacity of spherical condenser is
(A) $C = 4\pi\epsilon_0\left(\frac{rR}{r-R}\right)$ (B) $C = 4\pi\epsilon_0\left(\frac{rR^2}{r-R}\right)$ (C) $C = 4\pi\epsilon_0\left(\frac{rR}{r+R}\right)$ (D) $C = 4\pi\epsilon_0\left(\frac{r^2R}{r-R}\right)$

62. Capacitors of 3 µF and 6 µF are connected in series. Their equivalent capacitance is:
(A) 3 µF (B) 18 µF (C) 9 µF (D) 2 µF

63. The capacity of any conductor is
(A) $C = q \cdot v$ (B) $C = q/v$ (C) $C = v/q$ (D) $C = \frac{1}{2}q \cdot v$

64. Energy of a charged conductor is
(A) $E = C \cdot V$ (B) $E = \frac{1}{2}C \cdot V$ (C) $E = \frac{1}{2}C^2V$ (D) $E = \frac{1}{2}CV^2$

65. A 4 µF capacitor connected to a 12 V battery carries charge:
(A) 48 µC (B) 3 µC (C) 0.33 µC (D) 16 µC

66. If n capacitors of equal capacity C_1 are connected in parallel, the equivalent capacity will be
(A) $C = n^2C_1$ (B) $C = n/C_1$ (C) $C = nC_1$ (D) $C = C_1/n$

67. Dielectric strength for air is
(A) 4×10^6 Vm^{-1} (B) 10^6 Vm^{-1} (C) 3×10^6 Vm^{-1} (D) 5×10^6 Vm^{-1}

68. A 5 µF capacitor carries charge 2 µC. The potential difference across it is:
(A) 2.5 V (B) 0.4 V (C) 10 V (D) 25 V

69. Van de Graaff generator is an electrostatic machine which produces
(A) Only high voltage (B) Only high current (C) Low current and low voltage (D) High current and high voltage

70. The net charge on a charged capacitor is
(A) 1 C (B) infinite (C) $1\mu C$ (D) zero

71. The correct expression for energy stored in a capacitor is:
(A) $2CV^2$ (B) Q^2/C (C) QV (D) $Q^2/(2C)$

72. Three capacitors of capacitance $6\mu F$ are available. The minimum and maximum capacitances obtained are
(A) $2\mu F, 12\mu F$ (B) $6\mu F, 18\mu F$ (C) $2\mu F, 18\mu F$ (D) $3\mu F, 12\mu F$

73. The energy density in a parallel-plate capacitor is:
(A) $\frac{1}{2}\epsilon_0 E^2$ (B) $\epsilon_0 E^2$ (C) $2\epsilon_0 E$ (D) $\frac{1}{2}\epsilon_0/E^2$

74. The value of electric potential at a distance r from a point charge is
(A) inversely proportional to r^2 (B) proportional to r (C) proportional to r^2 (D) inversely proportional to r

75. 64 identical drops each of capacity $5\mu F$ combine to form a big drop. What will be the capacity of the big drop?
(A) $164\mu F$ (B) $20\mu F$ (C) $25\mu F$ (D) $4\mu F$

76. When a dielectric completely fills a capacitor that remains connected to a battery, which quantity stays constant?
(A) charge (B) stored energy (C) capacitance (D) potential difference

77. If the distance between the two charges is increased, then the electrostatic potential energy of the charges
(A) remains the same (B) increases (C) may increase or decreases (D) decrease

78. In broad-side-on position, the electric potential due to electric dipole is
(A) zero (B) infinite (C) $\frac{1}{4\pi\epsilon_0}\frac{p}{r^2}$ (D) $\frac{1}{4\pi\epsilon_0}\frac{p}{r}$

79. When a dielectric is inserted into a capacitor disconnected from a battery, which quantity stays constant?
(A) capacitance (B) energy density (C) potential difference (D) charge

80. An electron is accelerated to 5 volt potential difference. The energy gained by the electron will be
(A) 5 joule (B) 5 watt (C) 5 erg (D) 5 eV

81. The relation between electric field (E) and electric potential (V) is
(A) $E = -\frac{dr}{dV}$ (B) $E = \frac{dV}{dr}$ (C) $E = -\frac{dV}{dr}$ (D) $E = \frac{dr}{dV}$

82. When a dielectric of constant K completely fills a capacitor, its capacitance:
(A) becomes K times (B) becomes zero (C) is divided by K (D) remains unchanged

83. The electrostatic energy of the system made by two electric dipoles kept at a distance 'r' is proportional to
(A) r^4 (B) r^3 (C) r^2 (D) none of these

84. Picofarad is the unit of
(A) electric charge (B) electric flux (C) intensity of electric field (D) electric capacity

85. If the capacitance of a battery-connected capacitor is increased, the charge drawn from the battery:
(A) decreases (B) remains unchanged (C) increases (D) becomes zero

86. Capacity of any condenser does not depend upon
(A) size of plates (B) shape of plates (C) charges on plates (D) distance between plates

87. The capacity of a spherical conductor is $1.0\mu F$. Its radius will be
(A) 9 km (B) 1.11 cm (C) 10 metre (D) 1.11 metre

88. The dependence of the far-field potential on the axial line of an electric dipole is:
(A) $1/r^3$ (B) r^2 (C) $1/r^2$ (D) $1/r$

89. The dielectric constant of a metal is
(A) -1 (B) infinity (C) 1 (D) 0

90. Outside a uniformly charged spherical conductor, the potential varies as if all charge were:
(A) zero in the volume (B) concentrated at the centre (C) at equal distance on the surface (D) at infinity

91. Which of the following is blocked by a capacitor ?
(A) DC (B) Both AC and DC (C) Neither AC nor DC (D) AC

92. The angle between equipotential surface and electric lines of force is
(A) 45° (B) 180° (C) Zero (D) 90°

93. Inside a charged spherical conductor of radius R, with zero potential at infinity, the potential is:
(A) infinite at the centre (B) equal to the surface potential (C) proportional to radius (D) zero

94. Potential at distance X out side a charged sphere of radius R charged with charge Q is
(A) $\frac{kQ}{X}$ (B) $\frac{kQ}{X^2}$ (C) kXQ (D) $\frac{kQ}{R}$

95. An α-particle is accelerated by a potential difference 200 V. Increase in its energy will be
(A) 800 eV (B) 400 eV (C) 100 eV (D) Zero

96. The magnitude of electric field at distance r from a point charge q is:
(A) kq^2/r (B) kq/r (C) kr^2/q (D) kq/r^2

97. What is the energy stored in a capacitor of capacity C when its potential is increased up to V.
(A) $\frac{1}{2}CV$ (B) CV (C) $\frac{1}{2}CV^2$ (D) $\frac{1}{2CV}$

98. N equal spherical drops of equal Potential V are Combined to form a big drop. Potential of new big drop will be
(A) $V \times N$ (B) V (C) $\frac{V}{N}$ (D) $V \times N^{\frac{2}{3}}$

99. At the midpoint between two equal positive charges, the electric potential is:
(A) positive (B) negative (C) always zero irrespective of the charges (D) zero

100. Electric field in between the plates of a parallel plate condenser is 10^5 V/m. If charge on plates of condenser is 1 μc then force on each plate of condenser is
(A) None (B) 0.05 N (C) 0.5 N (D) 0.005 N